Ryerson University

Department of Mechanical and Industrial Engineering

MTL 200: Materials Science Fundamentals

Term Test 1 – Arkham Edition (Closed Book)

Instructor: Alan Machin

February 23, 2016

Duration: 50 minutes

Name: (First Name) [redacted] **(Last Name)** [redacted]

Section: 2

Student Number: [redacted]

Question	1	2	3	4	Total
Max. Marks	10	10	10	10	40
Marks Obtained	7	10	10	6.5	33.5

Question 1: Multiple Choice (Only one correct or most appropriate answer) **(1 mark each)**

1. Rank the types of secondary bonding in order from weakest to strongest:
   A. Polar molecule induced dipole bonding
   B. Hydrogen bonding
   C. Permanent dipole bonding
   D. Fluctuating induced dipole bonding

   a) D, A, B, C
   b) B, C, A, D
   c) C, B, D, A
   d) D, A, C, B

2. Which material would you expect to have the lowest thermal expansion coefficient?
   ($X_{Zn} = 1.6$, $X_O = 3.5$, $X_{Al} = 1.5$, $X_{Mg} = 1.2$, $X_H = 2.1$)
   a) ZnO 0.59
   b) Al_2O_3 0.63
   c) MgO 0.73 strong bonds high
   d) H_2O 0.39

   m/C°

3. If the bonding energy in a material is high, which property would you expect to see?
   a) Low strength
   b) Low ductility
   c) Low melting point
   d) Low density

4. What type of interatomic bonding is predominant for ZnO?
   a) Polar induced dipole bonding
   b) Covalent bonding
   c) Ionic bonding
   d) Metallic bonding

5. What is the coordination number of an FCC crystal structure?
   a) 4
   b) 6
   c) 8
   d) 12

6. An elemental solid that can exist with more than one crystal structure is an example of:
   a) Allotropy
   b) Amorphous
   c) Crystal system
   d) Anisotropy

7. In general, which crystal structure do you expect to be the most brittle?
   a) BCC
   b) FCC
   c) HCP
   d) BAT

8. Appreciable quantities of a solute may be accommodated in a substitutional solid solution when the difference in the atomic radii between the two atoms is less than:
   a) ±5%
   b) ±10%
   c) ±15%
   d) ±20%

9. One would predict Co to dissolve ___less (1)___ than Cr in Fe, and Ni to dissolve ___more (2)___ than Al in Ag.
   a) (1) more, (2) more
   b) (1) less, (2) less
   c) (1) more, (2) less
   d) (1) less, (2) more

10. Which condition is NOT true: Diffusion is slower for:
   a) Materials with covalent bonding rather than secondary bonding
   b) Smaller diffusing atoms
   c) Closer-packed structures
   d) Higher density materials

$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$

Question 2: Short Answer. (Enter the final answer in the box provided. Only the final answer will be graded. Please include units in your answer if necessary, and reduce to proper form where applicable.)

(a) What is the electronic configuration for Ti^{1+} ($Z_{Ti} = 22$): **(3 marks)**

$Ti = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$

Ti^{1+} $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

1 s
2 s p
3 s p d
4 s p d f
5 s p

Answer: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$ 3

(b) If you counted 25 grains/in^2 on a micrograph at 100x magnification, how many grains per square inch would you expect to see on the same sample at 200x magnification? **(3 marks)**

$N_m = 2^{n-1}$

$25 = 2^{n-1}$

$\frac{\ln 25}{\ln 2} + 1 = n$

$n = 5.64$

$N_m \left(\frac{200}{100}\right)^2 = 2^{n-1}$

$N_m = 2^{5.64-1} \div \left(\frac{200}{100}\right)^2$

$= 6.25$ grain/in^2

Answer: 6.25 grain/in^2 3

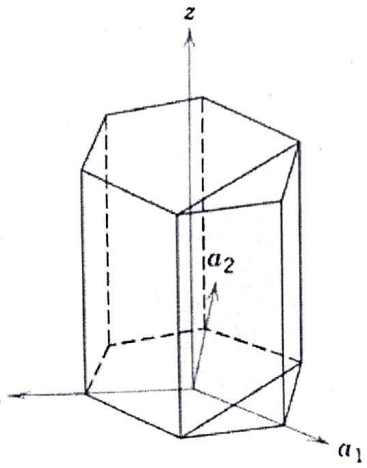
(c) Determine the Miller-Bravais (hexagonal) or Miller (cubic) index for the following crystallographic planes. **(2 marks each)**

$a_1 = \frac{1}{\frac{1}{2}} = 2$

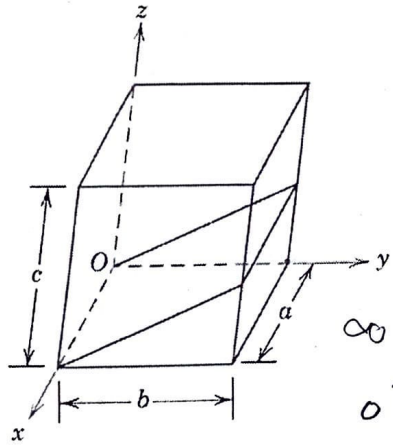
$a_2 = \frac{-\frac{1}{2}}{1} = -1$

$a_3 = \frac{-1}{1} = -1$

$a_4 = \frac{1}{\infty} = 0$



$(2\bar{1}\bar{1}0)$ 2



$\infty\ \bar{1}\ \frac{1}{2}$

$0\bar{1}2$

$(0\bar{1}2)$ 2

Question 3: Commissioner Gordon has been kidnapped! Batman found an unknown metallic substance at Gordon's last known location. By subjecting the material to first-order monochromatic x-ray diffraction using a wavelength of 1.5405 Å, Batman noticed that a diffraction angle of 65.025° was found for the (211) plane. Assuming the unknown material has a BCC structure, from the table below, which criminal should Batman pursue? **(10 marks)**

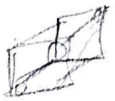
Criminal	Signature Material	Atomic Radius (Å)
Riddler	Lithium	1.52
Mr Freeze	Cesium	2.65
Penguin	Chromium	1.25
Bane	Barium	2.17

$\lambda = 2d \sin\theta^\circ$

$d = \dfrac{\lambda}{2\sin\theta}$

$= \dfrac{1.5405\ A^\circ}{2\sin\left(\frac{65.025^\circ}{2}\right)}$

$= 1.433\ A$



$(4r)^2 = a^2 + 2a^2$

$(4r)^2 = 3a^2$

$4r = \sqrt{3}a$

$a = \dfrac{4}{\sqrt{3}} r$

$d = \dfrac{a}{\sqrt{h^2+k^2+l^2}}$

$1.433A^\circ = \dfrac{\frac{4}{\sqrt{3}} r}{\sqrt{2^2+1^2+1^2}}$

$r = 1.52\ A^\circ$

$\therefore$ Atomic radius match that of Lithium

$\therefore$ Batman should pursue Riddler.

Question 4: Joker has armed a bomb that is set to explode in 15 h. The bomb is impervious to Batman's typical batarangs, but he knows that through case hardening the pure iron batarangs in gaseous nitrogen, he can produce a strong enough exterior to penetrate the bomb's defenses: 0.05 wt.% N at a position 2 mm from the surface. Due to material constraints, Batman can only treat the batarangs at a maximum temperature of 675 °C. The diffusion coefficient for nitrogen in iron at 675 °C is 2.8 x 10^{-11} m^2/s. Maintaining a surface concentration of 0.2 wt.% N:

(a) How long will it take to case harden the batarangs as required? **(9 marks)**
(b) Will he be able to save the people of Gotham? **(1 mark)**

a)

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\frac{0.05 - 0}{0.2 - 0} = 1 - \text{erf}\left(\frac{2 \times 10^{-3}}{2\sqrt{2.8 \times 10^{-11} \times t}}\right)$$

$$0.75 = \text{erf}\left(\frac{2 \times 10^{-3}}{2\sqrt{2.8 \times 10^{-11} \times t}}\right)$$

erf (z) z

0.75 z_x

$z_x = 0.7112$

$$0.7112 = \frac{2 \times 10^{-3}}{2\sqrt{2.8 \times 10^{-11} \times t}}$$

$$1.4224\sqrt{2.8 \times 10^{-11} \times t} = 2 \times 10^{-3}$$

$$\sqrt{2.8 \times 10^{-11} \times t} = 1.406 \times 10^{-3}$$

$$t = 70608.7418$$

$$= 19.6h$$

$$\approx \underline{20h}$$

b) ∴ he will not be able to save people of Gotham

Formula Sheet

$$\%IC = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100$$

$$n\lambda = 2d\sin\theta$$

$$N_M \left(\frac{M}{100}\right)^2 = 2^{n-1}$$

$$D = D_0 e^{-\left(\frac{Q_d}{RT}\right)}$$

$$J = -D\frac{dC}{dx} = -D\frac{C_A - C_B}{x_A - x_B}$$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - erf\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

z	*erf(z)*	*z*	*erf(z)*	*z*	*erf(z)*
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Element	*Atomic Radius (nm)*	*Crystal Structure*	*Electro-negativity*	*Valence*
Ni	0.1246	FCC	1.8	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Pt	0.1387	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2